

# **DIGITAL PSYCHROMETRIC CHART**

Numerical Analysis (MATLAB) Project

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## **Abstract**

In this project we have developed a program that can calculate air properties for a given dry and wet bulb temperature using psychrometric calculations. As it is an important tool for engineers and engineering students, this program will help them who deal with psychrometric chart always and for applying the values for various projects or needs. Using the psychrometric chart is pesky and evasive.

## **Acknowledgement**

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## I. Introduction

This project is about digitalizing the psychrometric chart which is tedious to calculate manually. It will reduce the workload of mechanical engineers who use psychrometric chart every now and then. Our main view is to calculate all the air properties in a much more accurate way than the orthodox process. These air properties are very much important to know about the intended condition of environment.

As it is a program, it can be used in any digital device that can measure different air properties. Air Conditioners, Heaters, Industrial dryers, Washing Machines can manipulate temperature and humidity to get the required conditions. Also in cases like bigger malls, fully conditioned environments, multi storied conditioning systems, this can automatically calculate and make decisions to change the environment to a comfort or required zone.

So it will be helpful for various purposes and it can also be used industrially in a macro level.

## II. Problem Specifications

Different air properties like ‘relative humidity’, ‘humidity ratio’, ‘specific volume’, ‘enthalpy’ etc. can be measured using only dry and wet bulb temperature.

Here dry and wet bulb temperature are our input variables, and the output will be several other psychrometric properties up to a certain accuracy limit. It can calculate from 0 degree dry bulb temperature to 200 degree dry bulb temperature. This range can be expanded if the file called in the code is enriched with more saturation pressure and temperature.

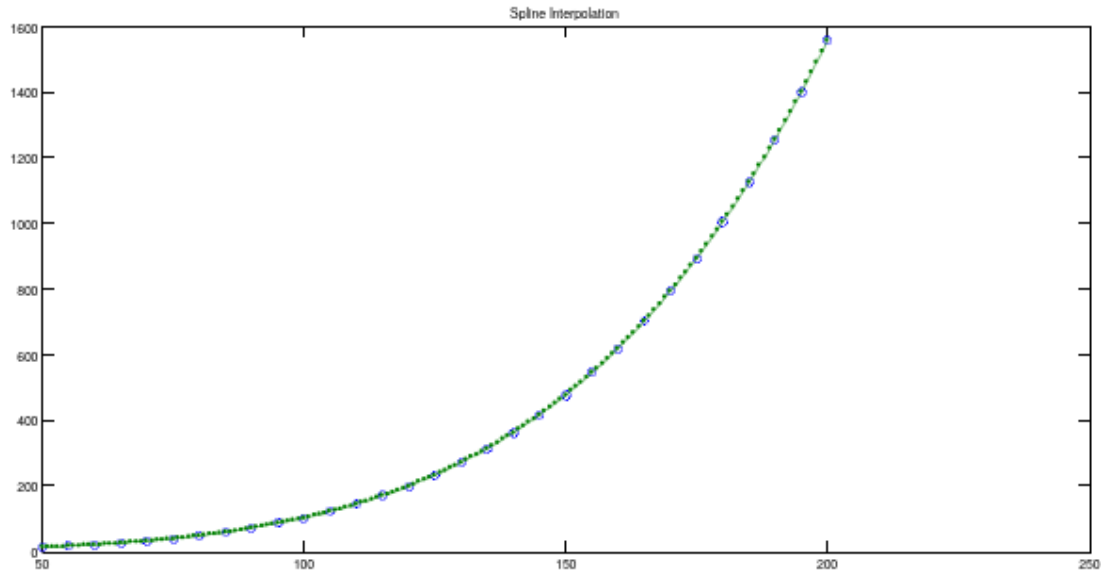
For example, say the dry bulb temperature and wet bulb temperature is 36 degree Celsius and 30 degree Celsius respectively. Then the program will show as following (see Fig. 2).

## III. Nomenclature

|           |                                     |
|-----------|-------------------------------------|
| $p_{g1}$  | Saturated pressure at a temperature |
| $p_{atm}$ | Atmospheric pressure                |
| $h$       | Enthalpy                            |
| $h_r$     | Humidity ratio                      |
| $rh$      | Relative humidity                   |
| $vol$     | Specific volume                     |

## IV. Methodology

A file input is used for a set of temperature versus saturation pressure data is given. We used “Thermodynamics: An Engineering Approach” by Yunus A. Cengel for these sets of data. Later it was interpolated for more accurate values.



*Fig. 1: Spline interpolation and rapid change of saturation pressure.*

Code works on the equations stated below.

$$wg1 = 621.98 * pg1 ./ (patm - pg1);$$

It calculates a temporary humidity ratio which would only be true for equal dry and wet bulb temperature.

$$h = t1 + 2.5 * wg1;$$

Then enthalpy is calculated using this equation.

$$\begin{aligned} t0 &= h; \\ hr &= (db - t1) ./ (t1 - t0) * wg1 + wg1; \end{aligned}$$

Actual humidity ratio is calculated.

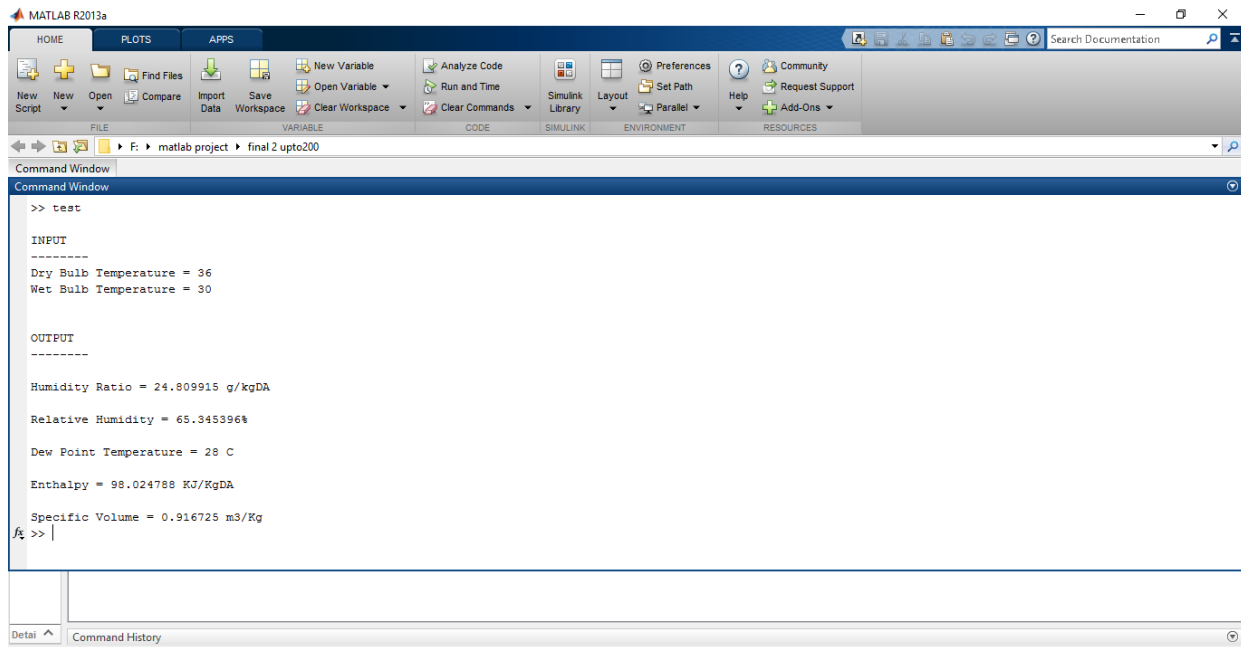
$$\begin{aligned} prs &= (hr * 101.325) / (621.98 + hr); \\ rh &= prs / trh * 100; \end{aligned}$$

Respective relative humidity is determined.

$$vol = rair * (t1 + 273) ./ (patm - pg1);$$

Specific volume of air is calculated using saturated pressure and temperature.

## V. Results



*Fig. 2(a): Program input and output in the MATLAB command window.*

```
>> test

INPUT
-----

Dry Bulb Temperature = 36
Wet Bulb Temperature = 30

OUTPUT
-----

Humidity Ratio = 24.809915 g/kgDA

Relative Humidity = 65.345396%

Dew Point Temperature = 28 C

Enthalpy = 98.024788 KJ/KgDA

Specific Volume = 0.916725 m3/Kg
fx >> |
```

*Fig. 2(b): Results.*

project\_psy\_gui

Dry Bulb Temp. 26 °C

Wet Bulb Temp. 23 °C

Calculate

Relative Humidity 78.063 %

Humidity Ratio 16.5482 Kg/KgDA

Dew Point Temp. 21 °C

Specific Volume 0.86585 m3/Kg

Specific Enthalpy 67.3706 KJ/KgDA

Fig. 3: Graphic user interface.

|      |         |       |             |        |              |        |              |        |               |        |               |
|------|---------|-------|-------------|--------|--------------|--------|--------------|--------|---------------|--------|---------------|
| p.01 | 0.61165 | 37    | 6.28230     | 74.00  | 37.01033459  | 111.00 | 148.26317933 | 147.00 | 439.08409501  | 165.00 | 700.93000000  |
| 1    | 0.65709 | 38    | 6.63280     | 75.00  | 38.59700000  | 112.00 | 153.28163327 | 148.00 | 451.17600065  | 166.00 | 718.47600829  |
| 2    | 0.70599 | 39    | 7.00020     | 76.00  | 40.24050087  | 113.00 | 158.43849755 | 149.00 | 463.53327087  | 167.00 | 736.36881154  |
| 3    | 0.75808 | 40    | 7.38490     | 77.00  | 41.94247869  | 114.00 | 163.73690788 | 150.00 | 476.16000000  | 168.00 | 754.61361066  |
| 4    | 0.81355 | 41    | 7.78780     | 78.00  | 43.70460607  | 115.00 | 169.18000000 | 151.00 | 489.06037175  | 169.00 | 773.21560652  |
| 5    | 0.87258 | 42    | 8.20960     | 79.00  | 45.52855563  | 116.00 | 174.77086612 | 152.00 | 502.23892720  | 170.00 | 792.18000000  |
| 6    | 0.93536 | 43    | 8.65080     | 80.00  | 47.41600000  | 117.00 | 180.51242445 | 153.00 | 515.70029678  | 171.00 | 811.51183229  |
| 7    | 1.00210 | 44    | 9.11240     | 81.00  | 49.36864230  | 118.00 | 186.40754973 | 154.00 | 529.44911090  | 172.00 | 831.21550575  |
| 8    | 1.07300 | 45    | 9.59500     | 82.00  | 51.38830774  | 119.00 | 192.45911667 | 155.00 | 543.49000000  | 173.00 | 851.29526308  |
| 9    | 1.14830 | 46    | 10.09900    | 83.00  | 53.47685203  | 120.00 | 198.67000000 | 156.00 | 557.82753342  | 174.00 | 871.75534693  |
| 10   | 1.22820 | 47    | 10.62700    | 84.00  | 55.63613088  | 121.00 | 205.04319621 | 157.00 | 572.46603618  | 175.00 | 892.60000000  |
| 11   | 1.31300 | 48    | 11.17700    | 85.00  | 57.86800000  | 122.00 | 211.58218892 | 158.00 | 587.40977224  | 176.00 | 913.83394257  |
| 12   | 1.40280 | 49    | 11.75200    | 86.00  | 60.17432991  | 123.00 | 218.29058352 | 159.00 | 602.66300553  | 177.00 | 935.46380544  |
| 13   | 1.49810 | 50    | 12.35200    | 87.00  | 62.55705034  | 124.00 | 225.17198542 | 160.00 | 618.23000000  | 178.00 | 957.49669703  |
| 14   | 1.59900 | 51.00 | 12.97894362 | 88.00  | 65.01810581  | 125.00 | 232.23000000 | 161.00 | 634.11509457  | 179.00 | 979.93972574  |
| 15   | 1.70580 | 52.00 | 13.63241816 | 89.00  | 67.55944085  | 126.00 | 239.46810905 | 162.00 | 650.32292807  | 180.00 | 1002.80000000 |
| 16   | 1.81880 | 53.00 | 14.31352089 | 90.00  | 70.18300000  | 127.00 | 246.88929988 | 163.00 | 666.85821428  | 181.00 | 1026.08375743 |
| 17   | 1.93840 | 54.00 | 15.02334908 | 91.00  | 72.89078205  | 128.00 | 254.49643619 | 164.00 | 683.72566699  | 182.00 | 1049.79375248 |
| 18   | 2.06470 | 55.00 | 15.76300000 | 92.00  | 75.68500290  | 129.00 | 262.29238166 | 165.00 | 700.93000000  | 183.00 | 1073.93186881 |
| 19   | 2.19830 | 56.00 | 16.53357092 | 93.00  | 78.56793273  | 130.00 | 270.28000000 | 166.00 | 718.47600829  | 184.00 | 1098.49999010 |
| 20   | 2.33930 | 57.00 | 17.33615911 | 94.00  | 81.54184170  | 131.00 | 278.46236758 | 167.00 | 736.36881154  | 185.00 | 1123.50000000 |
| 21   | 2.48820 | 58.00 | 18.17186184 | 95.00  | 84.60900000  | 132.00 | 286.84341155 | 168.00 | 754.61361066  | 186.00 | 1148.93502772 |
| 22   | 2.64530 | 59.00 | 19.04177638 | 96.00  | 87.77165389  | 133.00 | 295.42727172 | 169.00 | 773.21560652  | 187.00 | 1174.81318465 |
| 23   | 2.81110 | 60.00 | 19.94700000 | 97.00  | 91.03195406  | 134.00 | 304.21808793 | 170.00 | 792.18000000  | 188.00 | 1201.14382772 |
| 24   | 2.98580 | 61.00 | 20.88864470 | 98.00  | 94.39202728  | 135.00 | 313.22000000 | 171.00 | 811.51183229  | 189.00 | 1227.93631386 |
| 25   | 3.16990 | 62.00 | 21.86788141 | 99.00  | 97.85400033  | 136.00 | 322.43706062 | 172.00 | 831.21550575  | 190.00 | 1255.20000000 |
| 26   | 3.36390 | 63.00 | 22.88589576 | 100.00 | 101.42000000 | 137.00 | 331.87297392 | 173.00 | 851.29526308  | 191.00 | 1282.94333168 |
| 27   | 3.56810 | 64.00 | 23.94387341 | 101.00 | 105.09227438 | 138.00 | 341.53135692 | 174.00 | 871.75534693  | 192.00 | 1311.17110891 |
| 28   | 3.78310 | 65.00 | 25.04300000 | 102.00 | 108.87355687 | 139.00 | 351.41582661 | 175.00 | 892.60000000  | 193.00 | 1339.88722030 |
| 29   | 4.00920 | 66.00 | 26.18449827 | 103.00 | 112.76670216 | 140.00 | 361.53000000 | 176.00 | 913.83394257  | 194.00 | 1369.09555446 |
| 30   | 4.24690 | 67.00 | 27.36973927 | 104.00 | 116.77456497 | 141.00 | 371.87754995 | 177.00 | 935.46380544  | 195.00 | 1398.80000000 |
| 31   | 4.49690 | 68.00 | 28.60013114 | 105.00 | 120.90000000 | 142.00 | 382.46237277 | 178.00 | 957.49669703  | 196.00 | 1429.00444554 |
| 32   | 4.75960 | 69.00 | 29.87708200 | 106.00 | 125.14577657 | 143.00 | 393.28842061 | 179.00 | 979.93972574  | 197.00 | 1459.71277970 |
| 33   | 5.03540 | 70.00 | 31.20200000 | 107.00 | 129.51432247 | 144.00 | 404.35964563 | 180.00 | 1002.80000000 | 198.00 | 1490.92889109 |
| 34   | 5.32510 | 71.00 | 32.57630621 | 108.00 | 134.00798007 | 145.00 | 415.68000000 | 181.00 | 1026.08375743 | 199.00 | 1522.65666832 |
| 35   | 5.62900 | 72.00 | 34.00147351 | 109.00 | 138.62909179 | 146.00 | 427.25345957 | 182.00 | 1049.79375248 | 200.00 | 1554.90000000 |
| 36   | 5.94790 | 73.00 | 35.47898770 | 110.00 | 143.38000000 | 147.00 | 439.08409501 | 183.00 | 1073.93186881 |        |               |

Fig. 4: Data file.

## VI. Conclusion

The outcome of this project is to introduce a digital psychrometric chart. It will be useful for digital devices that can calculate automatically the air properties and after that they can change the surrounding in such a way so that it will be turned to such a condition which is in comfort zone for human being. This calculator can calculate any value for dry bulb temperatures between 0–200 degree Celsius. If the dry bulb temperature exceeds 200 degree Celsius, then saturation pressure changes very rapidly and can cause some error. For temperature below than zero degree Celsius, we need to update the saturation pressure data (s.txt).

## VII. References

1. Thermodynamics: An Engineering Approach by Yunus A. Cengel & Michael A. Boles (7th edition), Chart A5.
2. [https://www.ohio.edu/mechanical/thermo/Applied/Chapt.7\\_11/Psychro\\_chart/psychro.html](https://www.ohio.edu/mechanical/thermo/Applied/Chapt.7_11/Psychro_chart/psychro.html)

## VIII. Appendix

### Interpolation

```
function [ t ] = fl(db)
file = dlmread('s.txt','\t');
if floor(db)~=db
    f_db=floor(db);
    c_db=ceil(db);
    f_p=file(f_db,2);
    c_p=file(c_db,2);
    n_p = f_p + ((c_p-f_p)/(c_db-f_db))*(db-f_db);
    t=n_p;
    %disp(trh);
    %disp(c_p);
    %disp(f_p);
else
    t=file(db+1,2);
end
end
```

### Main code

```
tpg = dlmread('s.txt','\t');
fprintf('\nINPUT\n-----\n');
db=input('Dry Bulb Temperature = ');
wb=input('Wet Bulb Temperature = ');

patm = 101.325;
rair = 0.287;
t1 = wb;

pg1=fl(wb);

wg1 = 621.98*pg1./(patm-pg1);
```



```

h = t1 + 2.5*wg1;
t0 = h;
hr = (db-t1)./(t1-t0)*wg1+wg1;

prs = (hr*101.325)/(621.98+hr);
trh=f1(db);

%trh = tpg(db+1,2); %pressure at dry bulb temp.
rh = prs/trh*100; %relative humidity

for i=1:200
    if prs < tpg(i,2)
        dpt=tpg(i,1)-1;
        break;
    end
end

%-----
vol = rair.*(t1+273)./(patm-pg1);
%disp(vol);

tv0 = (hr/2.7475)+db;

if tv0>6.9665 & tv0<9.87215
    vol = 0.7986;
elseif tv0>9.87215 & tv0<12.7778
    vol = 0.807125;

elseif tv0>12.7778 & tv0<15.8271
    vol = 0.8160;
elseif tv0>15.8271 & tv0<18.8764
    vol = 0.8251;

elseif tv0>18.8764 & tv0<22.1132
    vol = 0.83456;
elseif tv0>22.1132 & tv0<25.3500
    vol = 0.84455;

elseif tv0>25.3500 & tv0<28.83045
    vol = 0.85495;
elseif tv0>28.83045 & tv0<32.3109
    vol = 0.86585;

elseif tv0>32.3109 & tv0<36.1072
    vol = 0.87742;
elseif tv0>36.1072 & tv0<39.9035
    vol = 0.889675;

elseif tv0>39.9035 & tv0<44.1098
    vol = 0.902775;
elseif tv0>44.1098 & tv0<48.3161
    vol = 0.916725;
end

fprintf('\nDry Bulb Temperature = %d C\nWet Bulb Temperature = %d C\n\n',db,wb);
fprintf('\n\nOUTPUT\n-----\n');
fprintf('\nHumidity Ratio = %f g/kgDA \n\nRelative Humidity = %f%%\nDew Point
Temperature = %d C\n\n',hr,rh,dpt);
fprintf('Enthalpy = %f KJ/KgDA\n\nSpecific Volume = %f m3/Kg\n',h,vol);

```

## Interpolating data used from books

```
x = 50:5:200;
v = [12.352 15.763 19.947 25.043 31.202 38.597 47.416 57.868 70.183 84.609 101.42
120.90 143.38 169.18 198.67 232.23 270.28 313.22 361.53 415.68 476.16 543.49 618.23
700.93 792.18 892.6 1002.8 1123.5 1255.2 1398.8 1554.9];
xq = 50:1:200;
vq2 = interp1(x,v,xq,'spline');
plot(x,v,'o',xq,vq2,':');
xlim([50 250]);
title('Spline Interpolation');
%the values are appended to the file
```

## GUI code

```
function varargout = project_psy_gui(varargin)
% PROJECT_PSY_GUI MATLAB code for project_psy_gui.fig
%   PROJECT_PSY_GUI, by itself, creates a new PROJECT_PSY_GUI or raises the
existing
%   singleton*.
%
%   TTT = PROJECT_PSY_GUI returns the handle to a new PROJECT_PSY_GUI or the handle
to
%   the existing singleton*.
%
%   PROJECT_PSY_GUI('CALLBACK',hObject,eventData,handles,...) calls the local
function named CALLBACK in PROJECT_PSY_GUI.M with the given input arguments.
%
%   PROJECT_PSY_GUI('Property','Value',...) creates a new PROJECT_PSY_GUI or raises
the
%   existing singleton*. Starting from the left, property value pairs are
%   applied to the GUI before project_psy_gui_OpeningFcn gets called. An
%   unrecognized property name or invalid value makes property application
%   stop. All inputs are passed to project_psy_gui_OpeningFcn via varargin.
%
%   *See GUI Options on GUIDE's Tools menu. Choose "GUI allows only one
%   instance to run (singleton)".
%
% See also: GUIDE, GUIDATA, GUIHANDLES
% Edit the above text to modify the response to help project_psy_gui
% Last Modified by GUIDE v2.5 04-May-2016 17:33:09
% Begin initialization code - DO NOT EDIT
gui_Singleton = 1;
gui_State = struct('gui_Name',       mfilename, ...
                  'gui_Singleton',   gui_Singleton, ...
                  'gui_OpeningFcn', @project_psy_gui_OpeningFcn, ...
                  'gui_OutputFcn',  @project_psy_gui_OutputFcn, ...
                  'gui_LayoutFcn',  [] , ...
                  'gui_Callback',    []);
if nargin && ischar(varargin{1})
    gui_State.gui_Callback = str2func(varargin{1});
end

if nargout
    [varargout{1:nargout}] = gui_mainfcn(gui_State, varargin{:});
else
    gui_mainfcn(gui_State, varargin{:});
end
% End initialization code - DO NOT EDIT
```

```

% --- Executes just before project_psy_gui is made visible.
function project_psy_gui_OpeningFcn(hObject, eventdata, handles, varargin)
% This function has no output args, see OutputFcn.
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)
% varargin    command line arguments to project_psy_gui (see VARARGIN)

% Choose default command line output for project_psy_gui
handles.output = hObject;

% Update handles structure
guidata(hObject, handles);

% UIWAIT makes project_psy_gui wait for user response (see UIRESUME)
% uiwait(handles.figure1);

% --- Outputs from this function are returned to the command line.
function varargout = project_psy_gui_OutputFcn(hObject, eventdata, handles)
% varargout  cell array for returning output args (see VARARGOUT);
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)

% Get default command line output from handles structure
varargout{1} = handles.output;

function db_Callback(hObject, eventdata, handles)
% hObject    handle to db (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)

% Hints: get(hObject,'String') returns contents of db as text
%        str2double(get(hObject,'String')) returns contents of db as a double

% --- Executes during object creation, after setting all properties.
function db_CreateFcn(hObject, eventdata, handles)
% hObject    handle to db (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     empty - handles not created until after all CreateFcns called

% Hint: edit controls usually have a white background on Windows.
%       See ISPC and COMPUTER.
if ispc && isequal(get(hObject,'BackgroundColor'),
get(0,'defaultUicontrolBackgroundColor'))
    set(hObject,'BackgroundColor','white');
end

function wb_Callback(hObject, eventdata, handles)
% hObject    handle to wb (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB

```

```

% handles      structure with handles and user data (see GUIDATA)

% Hints: get(hObject,'String') returns contents of wb as text
%         str2double(get(hObject,'String')) returns contents of wb as a double

% --- Executes during object creation, after setting all properties.
function wb_CreateFcn(hObject, eventdata, handles)
% hObject      handle to wb (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      empty - handles not created until after all CreateFcns called

% Hint: edit controls usually have a white background on Windows.
%         See ISPC and COMPUTER.
if ispc && isequal(get(hObject,'BackgroundColor'),
get(0,'defaultUicontrolBackgroundColor'))
    set(hObject,'BackgroundColor','white');
end

% --- Executes on button press in pushbutton1.
function pushbutton1_Callback(hObject, eventdata, handles)
% hObject      handle to pushbutton1 (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      structure with handles and user data (see GUIDATA)

tpg = dlmread('t_pg.txt','\t');

db = str2double(get(handles.db,'string'));
wb = str2double(get(handles.wb,'string'));

patm = 101.325;
rair = 0.287;
t1 = wb;
pg1 = tpg(wb+1,2);
wg1 = 621.98*pg1./(patm-pg1);

h = t1 + 2.5*wg1;
t0 = h;
hr = (db-t1)./(t1-t0)*wg1+wg1;

prs = (hr*101.325)/(621.98+hr);

trh = tpg(db+1,2); %pressure at dry bulb temp.
rh = prs/trh*100; %relative humidity

for i=1:50
    if db == wb
        dpt = wb;
    elseif prs < tpg(i,2)
        dpt=tpg(i-1,1);
        disp(tpg(i,2));
        break;
    end
end
end

```

```

vol = rair.*(tl+273)./(patm-pgl);

tv0 = (hr/2.7475)+db;

if tv0>6.9665 & tv0<9.87215
    vol = 0.7986;
elseif tv0>9.87215 & tv0<12.7778
    vol = 0.807125;

elseif tv0>12.7778 & tv0<15.8271
    vol = 0.8160;
elseif tv0>15.8271 & tv0<18.8764
    vol = 0.8251;

elseif tv0>18.8764 & tv0<22.1132
    vol = 0.83456;
elseif tv0>22.1132 & tv0<25.3500
    vol = 0.84455;

elseif tv0>25.3500 & tv0<28.83045
    vol = 0.85495;
elseif tv0>28.83045 & tv0<32.3109
    vol = 0.86585;

elseif tv0>32.3109 & tv0<36.1072
    vol = 0.87742;
elseif tv0>36.1072 & tv0<39.9035
    vol = 0.889675;

elseif tv0>39.9035 & tv0<44.1098
    vol = 0.902775;
elseif tv0>44.1098 & tv0<48.3161
    vol = 0.916725;
end

hr = num2str(hr);
rh = num2str(rh);
vol = num2str(vol);
h = num2str(h);
dp = num2str(dp);

set(handles.hr,'string',hr);
set(handles.rh,'string',rh);
set(handles.vol,'string',vol);
set(handles.h,'string',h);
set(handles.dp,'string',dp);

```

## IX. Figures

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